

Electricidad I:

Respuesta completa de un circuito RC a un escalón

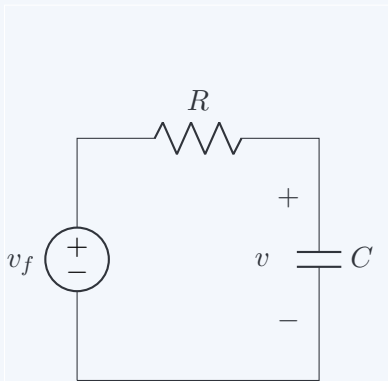
Juan J. Rojas

Instituto Tecnológico de Costa Rica
4 de octubre de 2026

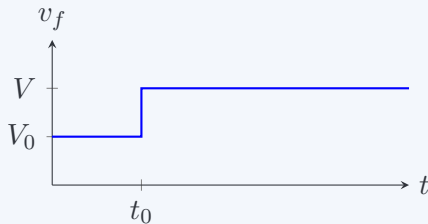
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de Costa Rica



Circuito RC: respuesta a un escalón



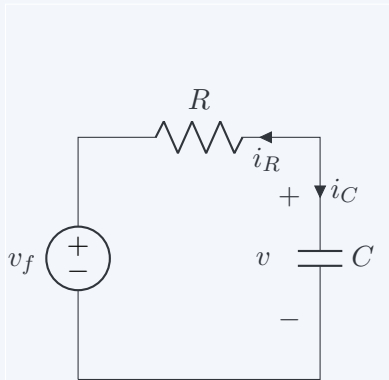
$$v_f(t) = V_0 u(t_0 - t) + V u(t - t_0)$$



Se asume que ha pasado mucho tiempo
y por lo tanto el capacitor tiene un
voltaje inicial

$$v(t_0) = V_0$$

Circuito RC: respuesta a un escalón

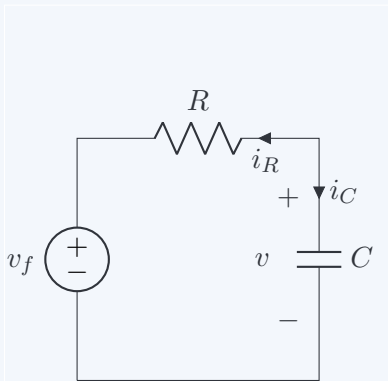


$$v_f(t) = V_0 u(t_0 - t) + V u(t - t_0)$$

aplicando LCK tenemos que:

$$i_C + i_R = 0$$

Circuito RC: respuesta a un escalón



$$v_f(t) = V_0 u(t_0 - t) + V u(t - t_0)$$

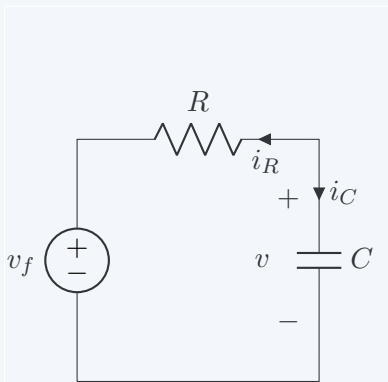
aplicando LCK tenemos que:

$$i_C + i_R = 0$$

y sabemos que para todo $t > t_0$:

$$i_C = C \frac{dv}{dt}, i_R = \frac{v - V}{R}$$

Circuito RC: respuesta a un escalón



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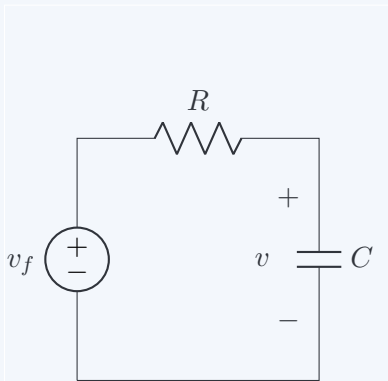
y sabemos que para todo $t > t_0$:

$$i_C = C \frac{dv}{dt}, i_R = \frac{v - V}{R}$$

entonces:

$$C \frac{dv}{dt} + \frac{v - V}{R} = 0$$

Circuito RC: respuesta a un escalón

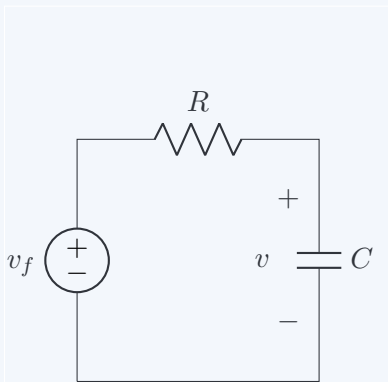


$$v_f(t) = V_0 u(t_0 - t) + V u(t - t_0)$$

dividiendo entre C :

$$\frac{dv}{dt} = -\frac{v - V}{RC}$$

Circuito RC: respuesta a un escalón



$$v_f(t) = V_0 u(t_0 - t) + V u(t - t_0)$$

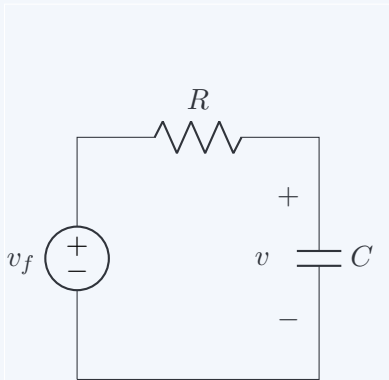
dividiendo entre C :

$$\frac{dv}{dt} = -\frac{v - V}{RC}$$

reordenado e integrando a ambos lados:

$$\int_{v(t_0)}^{v(t)} \frac{1}{v - V} dv = \int_{t_0}^t -\frac{1}{RC} dt$$

Circuito RC: respuesta a un escalón

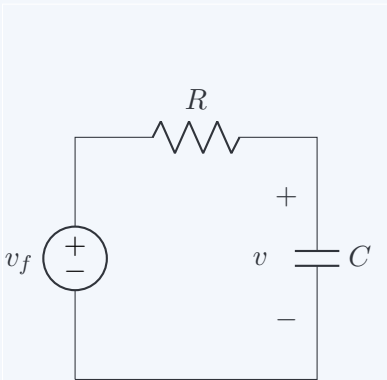


$$v_f(t) = V_0 u(t_0 - t) + V u(t - t_0)$$

tenemos que:

$$\ln |v - V| \bigg|_{v(t_0)}^{v(t)} = -\frac{1}{RC} t \bigg|_{t_0}^t$$

Circuito RC: respuesta a un escalón



$$v_f(t) = V_0 u(t_0 - t) + V u(t - t_0)$$

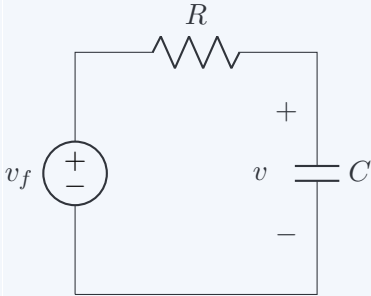
tenemos que:

$$\ln |v - V| \bigg|_{v(t_0)}^{v(t)} = -\frac{1}{RC} t \bigg|_{t_0}^t$$

evaluando:

$$\ln \left| \frac{v(t) - V}{v(t_0) - V} \right| = -\frac{1}{RC} (t - t_0)$$

Circuito RC: respuesta a un escalón

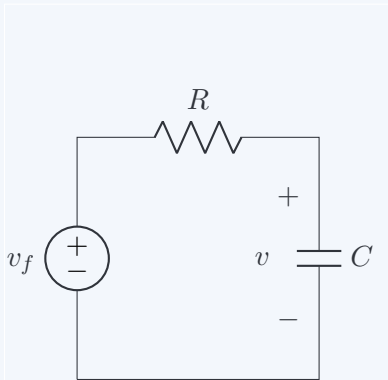


$$v_f(t) = V_0 u(t_0 - t) + V u(t - t_0)$$

aplicamos la función exponencial a
ambos lados

$$\frac{v(t) - V}{v(t_0) - V} = e^{-(t-t_0)/RC}$$

Circuito RC: respuesta a un escalón



$$v_f(t) = V_0 u(t_0 - t) + V u(t - t_0)$$

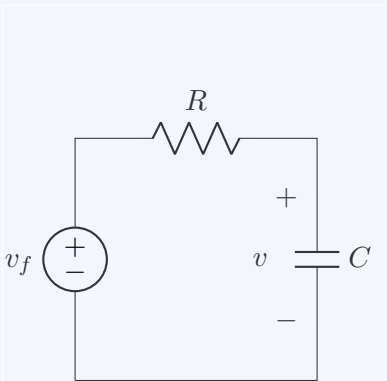
aplicamos la función exponencial a
ambos lados

$$\frac{v(t) - V}{v(t_0) - V} = e^{-(t-t_0)/RC}$$

sea $\tau = RC$ y $v(t_0) = V_0$, obtenemos:

$$v(t) = V + (V_0 - V)e^{-(t-t_0)/\tau}$$

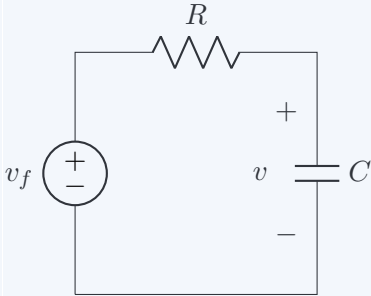
Circuito RC: respuesta a un escalón



$$v_f(t) = V_0 u(t_0 - t) + V u(t - t_0)$$

$$v(t) = \begin{cases} V_0, & t < t_0 \\ V + (V_0 - V)e^{-(t-t_0)/\tau}, & t > t_0 \end{cases}$$

Circuito RC: respuesta a un escalón

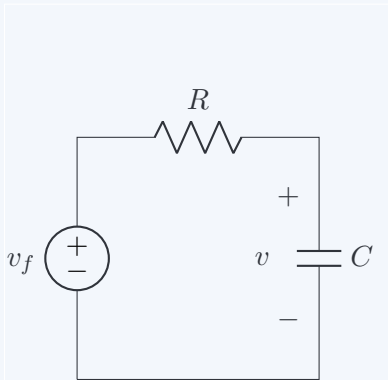


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$$i(t) = C \frac{dv}{dt}$$

Circuito RC: respuesta a un escalón



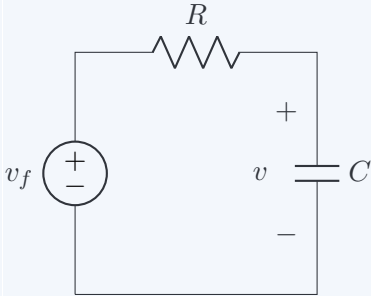
$$v_f(t) = V_0 u(t_0 - t) + V u(t - t_0)$$

$$v(t) = \begin{cases} V_0, & t < t_0 \\ V + (V_0 - V)e^{-(t-t_0)/\tau}, & t > t_0 \end{cases}$$

$$i(t) = C \frac{dv}{dt}$$

$$i(t) = \begin{cases} 0, & t < t_0 \\ \frac{(V - V_0)}{R} e^{-(t-t_0)/\tau}, & t > t_0 \end{cases}$$

Circuito RC: respuesta a un escalón

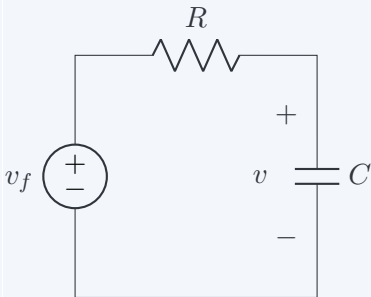


$$v_f(t) = V_0 u(t_0 - t) + V u(t - t_0)$$

Para $t > t_0$ y reordenando tenemos:

$$v(t) = V_0 e^{-(t-t_0)/\tau} + V(1 - e^{-(t-t_0)/\tau})$$

Circuito RC: respuesta a un escalón

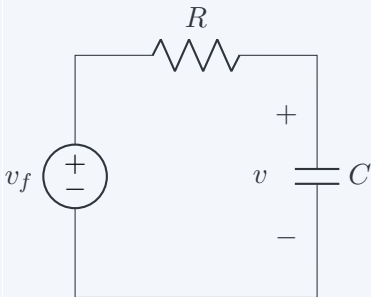


$$v_f(t) = V_0 u(t_0 - t) + V u(t - t_0)$$

Para $t > t_0$ y reordenando tenemos:

$$v(t) = \underbrace{V_0 e^{-(t-t_0)/\tau}}_{\text{Respuesta natural}} + V(1 - e^{-(t-t_0)/\tau})$$

Circuito RC: respuesta a un escalón

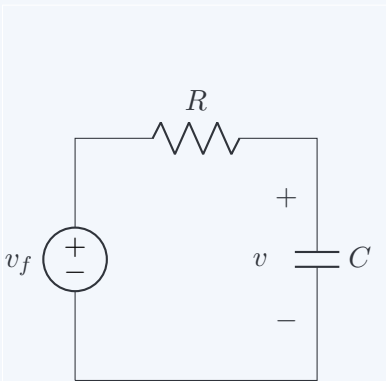


$$v_f(t) = V_0 u(t_0 - t) + V u(t - t_0)$$

Para $t > t_0$ y reordenando tenemos:

$$v(t) = \underbrace{V_0 e^{-(t-t_0)/\tau}}_{\text{Respuesta natural}} + \underbrace{V(1 - e^{-(t-t_0)/\tau})}_{\text{Respuesta forzada}}$$

Circuito RC: respuesta a un escalón

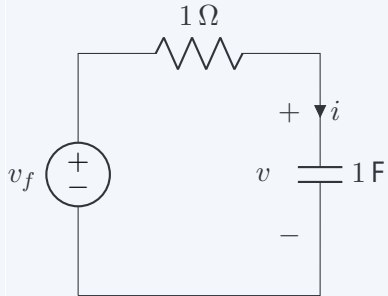


$$v_f(t) = V_0 u(t_0 - t) + V u(t - t_0)$$

Esta respuesta completa puede expresarse como:

$$v(t) = v(\infty) + [v(t_0) - v(\infty)]e^{-(t-t_0)/\tau}$$

Ejemplo

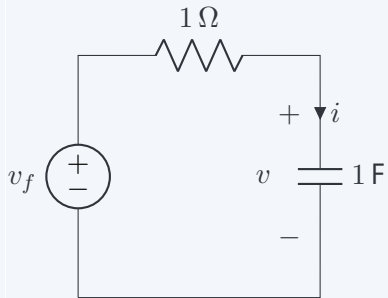


$$v_f(t) = 10\text{ V}[u(t - 1) - u(t - 3)]$$

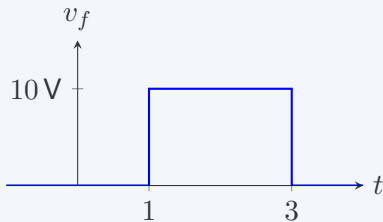
Objetivo:

Averiguar $v(t)$ e $i(t)$ para todo t

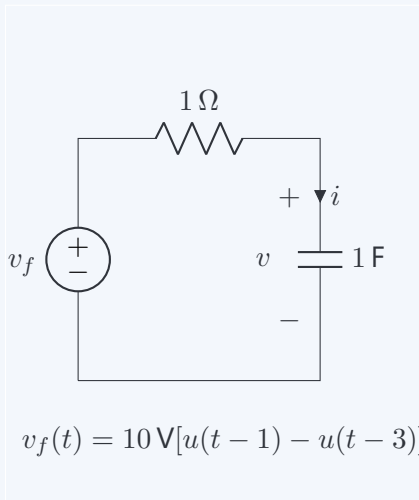
Ejemplo



$$v_f(t) = 10\text{ V}[u(t - 1) - u(t - 3)]$$



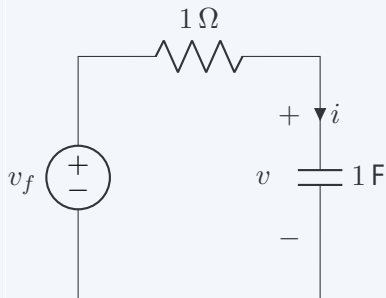
Ejemplo



para $t < 1$:

$$v(t) = 0\text{ V}$$

Ejemplo



$$v_f(t) = 10\text{ V}[u(t - 1) - u(t - 3)]$$

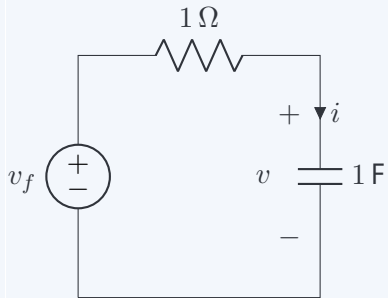
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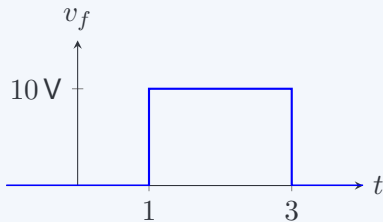
por lo tanto

$$v(1) = 0\text{ V}$$

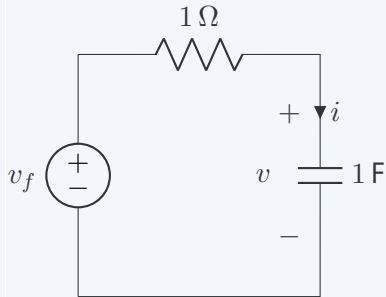
Ejemplo



$$v_f(t) = 10\text{ V}[u(t - 1) - u(t - 3)]$$



Ejemplo

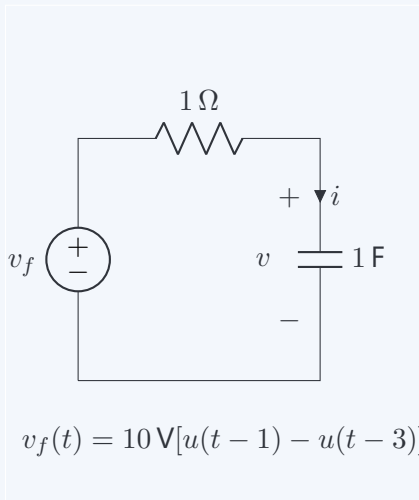


$$v_f(t) = 10\text{ V}[u(t - 1) - u(t - 3)]$$

para $1 \leq t < 3$:

$$v(t) = v(\infty) + [v(t_0) - v(\infty)]e^{-(t-t_0)/\tau}$$

Ejemplo



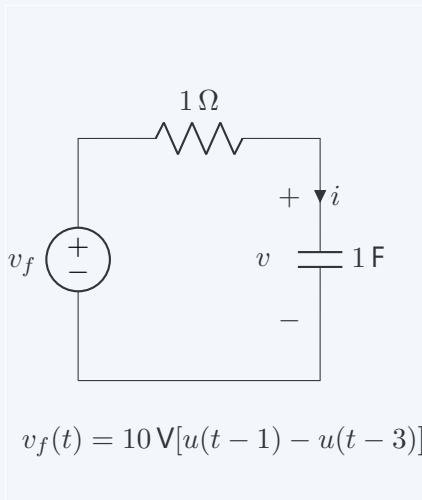
para $1 \leq t < 3$:

$$v(t) = v(\infty) + [v(t_0) - v(\infty)]e^{-(t-t_0)/\tau}$$

pero $v(\infty) = 10\text{ V}$, $\tau = 1\text{ s}$, $t_0 = 1\text{ s}$ y $v(1) = 0\text{ V}$, entonces:

$$v(t) = 10 \left(1 - e^{-(t-1)}\right)$$

Ejemplo



para $1 \leq t < 3$:

$$v(t) = v(\infty) + [v(t_0) - v(\infty)]e^{-(t-t_0)/\tau}$$

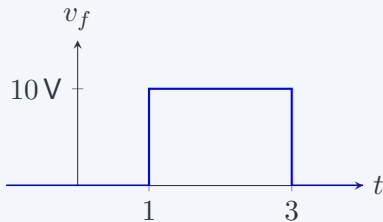
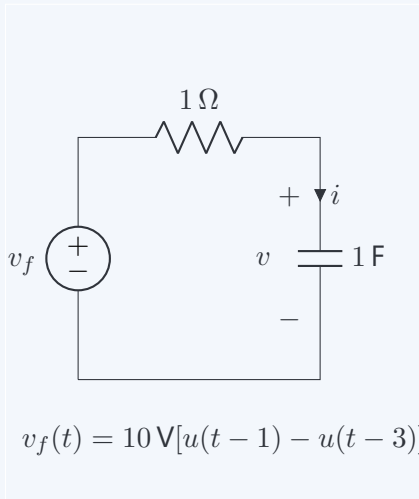
pero $v(\infty) = 10\text{ V}$, $\tau = 1\text{ s}$, $t_0 = 1\text{ s}$ y $v(1) = 0\text{ V}$, entonces:

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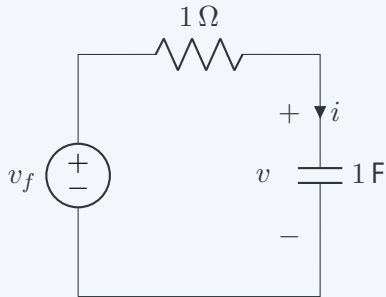
entonces:

$$v(3) = 10 \left(1 - e^{-2}\right) = 8,647\text{ V}$$

Ejemplo



Ejemplo

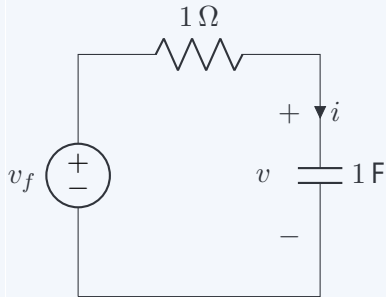


$$v_f(t) = 10\text{ V}[u(t - 1) - u(t - 3)]$$

para $t \geq 3$:

$$v(t) = v(\infty) + [v(t_0) - v(\infty)]e^{-(t-t_0)/\tau}$$

Ejemplo



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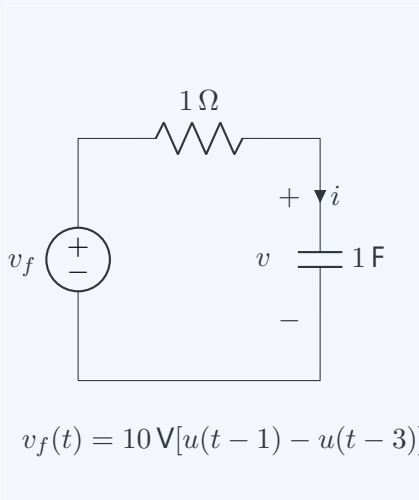
para $t \geq 3$:

$$v(t) = v(\infty) + [v(t_0) - v(\infty)]e^{-(t-t_0)/\tau}$$

pero $v(\infty) = 0\text{ V}$, $\tau = 1\text{ s}$, $t_0 = 3\text{ s}$ y
 $v(3) = 8,647\text{ V}$, entonces:

$$v(t) = 8,647e^{-(t-3)}$$

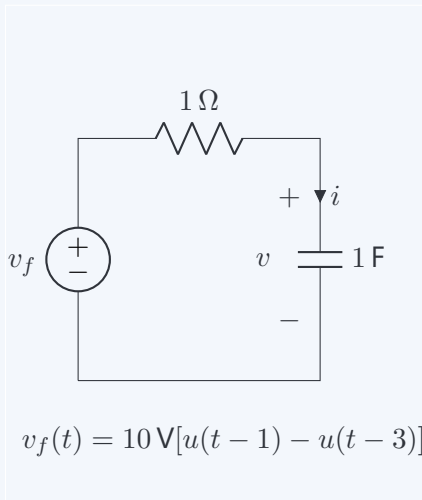
Ejemplo



Finalmente para todo t tenemos que:

$$v(t) = \begin{cases} 0 & \text{V, } t < 1 \\ 10(1 - e^{-(t-1)}) & \text{V, } 1 \leq t < 3 \\ 8,647e^{-(t-3)} & \text{V, } t \geq 3 \end{cases}$$

Ejemplo



Finalmente para todo t tenemos que:

$$v(t) = \begin{cases} 0 & \text{V, } t < 1 \\ 10(1 - e^{-(t-1)}) & \text{V, } 1 \leq t < 3 \\ 8,647e^{-(t-3)} & \text{V, } t \geq 3 \end{cases}$$

$$i(t) = \begin{cases} 0 & \text{A, } t < 1 \\ 10e^{-(t-1)} & \text{A, } 1 < t < 3 \\ -8,647e^{-(t-3)} & \text{A, } t > 3 \end{cases}$$

Ejemplo

